

ORF 309 Midterm 1 - Readable Cheat-sheet

1 Probability Foundations

1.1 Sample Space & Events

Sample space Ω : Set of all outcomes

Event: $A \subseteq \Omega$

Set operations:

- Union: $A \cup B$
- Intersection: $A \cap B$
- Complement: A^c
- Difference: $A \setminus B = A \cap B^c$

De Morgan's Laws:

- $(A \cup B)^c = A^c \cap B^c$
- $(A \cap B)^c = A^c \cup B^c$

1.2 Kolmogorov Axioms

1. $P(A) \geq 0$ (non-negative)
2. $P(\Omega) = 1$ (total probability)
3. $P(\bigcup A_i) = \sum P(A_i)$ for disjoint A_i (additivity)

Consequences:

- $P(\emptyset) = 0$
- $P(A^c) = 1 - P(A)$
- $A \subseteq B \Rightarrow P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Union Bound: $P(\bigcup A_i) \leq \sum P(A_i)$

1.3 Counting

Multiplication rule: k stages with n_i choices $\Rightarrow \prod n_i$ total

Permutations (ordered, no replacement):

$$P(n, k) = \frac{n!}{(n-k)!}$$

Combinations (unordered, no replacement):

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Multinomial:

$$\binom{n}{k_1, \dots, k_r} = \frac{n!}{k_1! \cdots k_r!}, \quad \sum k_i = n$$

Stars & bars: n balls into k bins:

$$\binom{n+k-1}{k-1}$$

Binomial theorem:

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Key identity: $\sum_{k=0}^n \binom{n}{k} = 2^n$

2 Conditional Probability

2.1 Basic Definitions

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Multiplication rule:

$$P(A \cap B) = P(A|B)P(B)$$

Chain rule:

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1) \prod_{i=2}^n P(A_i | \bigcap_{j<i} A_j)$$

2.2 Law of Total Probability (LOTP)

For partition $B_1, \dots, B_n$ (disjoint, $\bigcup B_i = \Omega$):

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

Two-event case:

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

2.3 Bayes' Rule

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}$$

Prior $P(B_i) \rightarrow$ **Posterior** $P(B_i|A)$ via **likelihood** $P(A|B_i)$

Sequential Bayes: Use posterior as new prior for next observation

2.4 Independence

Two events:

$$A \perp B \Leftrightarrow P(A \cap B) = P(A)P(B)$$

Equivalently: $P(A|B) = P(A)$ when $P(B) > 0$

If $A \perp B$: then $A \perp B^c$, $A^c \perp B$, $A^c \perp B^c$

Mutual independence: For every subset $S \subseteq \{1, \dots, n\}$, $|S| \geq 2$:

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i)$$

Pairwise $\nRightarrow$ mutual independence!

Conditional independence:

$$A \perp B|C \Leftrightarrow P(A \cap B|C) = P(A|C)P(B|C)$$

Does not imply unconditional independence (or vice versa)

IMPORTANT: Mutually exclusive $\neq$ independent!

- If $A \cap B = \emptyset$ and $P(A), P(B) > 0$: A, B are **dependent**

3 Discrete Random Variables

3.1 Definitions

RV: $X : \Omega \rightarrow \mathbb{R}$

PMF: $p_X(x) = P(X = x)$

- $p_X(x) \geq 0$
- $\sum_x p_X(x) = 1$

CDF: $F_X(x) = P(X \leq x)$

- Non-decreasing, right-continuous
- $F(-\infty) = 0$, $F(\infty) = 1$

$$P(a < X \leq b) = F(b) - F(a)$$

$$P(X > a) = 1 - F(a)$$

3.2 Expectation

$$E[X] = \sum_x x \cdot p_X(x)$$

LOTUS: $E[g(X)] = \sum_x g(x)p_X(x)$ (no need for PMF of $g(X)$)

Linearity (always, no independence needed):

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

Independence:

$$X \perp Y \Rightarrow E[XY] = E[X]E[Y]$$

(converse is false!)

3.3 Conditional Expectation

$$E[X|B] = \sum_x xP(X = x|B)$$

Tower / Law of Total Expectation:

$$E[X] = \sum_i E[X|B_i]P(B_i)$$

for partition $\{B_i\}$

$$E[X] = E[E[X|Y]]$$

3.4 Tail Sum Formula

For $X \in \{0, 1, 2, \dots\}$:

$$E[X] = \sum_{k=1}^{\infty} P(X \geq k) = \sum_{k=0}^{\infty} P(X > k)$$

3.5 Indicator Trick

Write $X = \sum 1_{A_i}$, then:

$$E[X] = \sum P(A_i)$$

Works even if dependent! For Var, need independence.

4 Named Discrete Distributions

4.1 Bernoulli(p)

$X \in \{0, 1\}$. $P(X = 1) = p$.

$$E[X] = p, \quad \text{Var} = p(1 - p)$$

4.2 Binomial(n, p)

$X = \#$ successes in n indep. Bern(p) trials

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, \dots, n$$

$$E[X] = np, \quad \text{Var} = np(1 - p)$$

Sum: $\text{Bin}(n, p) + \text{Bin}(m, p) = \text{Bin}(n + m, p)$

MGF: $(1 - p + pe^t)^n$

4.3 Geometric(p)

$T_1 =$ trial of 1st success. $T_1 \in \{1, 2, 3, \dots\}$

$$P(T_1 = k) = (1 - p)^{k-1} p$$

$$P(T_1 > k) = (1 - p)^k$$

$$E[T_1] = \frac{1}{p}, \quad \text{Var} = \frac{1 - p}{p^2}$$

Memoryless:

$$P(T_1 > m + n | T_1 > m) = P(T_1 > n)$$

Only discrete memoryless distribution.

MGF: $\frac{pe^t}{1 - (1 - p)e^t}$

4.4 Negative Binomial(r, p)

$T_r =$ trial of r -th success. $T_r \in \{r, r + 1, \dots\}$

$$P(T_r = n) = \binom{n-1}{r-1} p^r (1 - p)^{n-r}$$

$$E[T_r] = \frac{r}{p}, \quad \text{Var} = \frac{r(1 - p)}{p^2}$$

$T_r =$ sum of r i.i.d. Geom(p).

NegBin($1, p$) = Geom(p).

4.5 Poisson(λ)

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

$$E[X] = \lambda, \quad \text{Var} = \lambda$$

Poisson approximation: n large, p small, $\lambda = np$:

$$\text{Bin}(n, p) \approx \text{Pois}(\lambda)$$

Sum: $\text{Pois}(\lambda_1) + \text{Pois}(\lambda_2) = \text{Pois}(\lambda_1 + \lambda_2)$

Thinning: Mark each with prob p indep $\Rightarrow \text{Pois}(\lambda p)$

Splitting: Given $N(s + t) = n$:

$$N(s) \sim \text{Bin}(n, s/(s + t))$$

MGF: $e^{\lambda(e^t - 1)}$

4.6 Hypergeometric

Pop N : K success, $N - K$ fail. Draw n without replacement.

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

$$E[X] = \frac{nK}{N}$$

5 Variance & Covariance

5.1 Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$$

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Properties:

- $\text{Var}(X) \geq 0$; $= 0$ iff X is constant a.s.
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$

- $E[X^2] = \text{Var}(X) + (E[X])^2$ (key identity!)

Independent:

$$X \perp Y : \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

General:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

5.2 Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Properties:

- $\text{Cov}(X, X) = \text{Var}(X)$
- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$
- $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$ (bilinear)

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

(converse is false!)

5.3 Correlation

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}$$

$$-1 \leq \rho \leq 1$$

$$|\rho| = 1 \Leftrightarrow Y = aX + b \text{ (perfect linear)}$$

Uncorrelated ($\rho = 0$) $\neq$ independent

5.4 Variance of Linear Combo

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

Var via indicators: $X = \sum_i 1_{A_i}$ (indep Bern):

$$\text{Var}(X) = np(1 - p)$$

6 Continuous Random Variables

6.1 PDF & CDF

PDF: $f_X(x) \geq 0$, $\int_{-\infty}^{\infty} f_X(x) dx = 1$

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

$P(X = a) = 0$ for any single point

CDF: $F_X(x) = \int_{-\infty}^x f_X(t) dt$

$$f_X(x) = F'_X(x)$$

6.2 Expectation

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

LOTUS:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Variance:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) dx = E[X^2] - (E[X])^2$$

7 Named Continuous Distributions

7.1 Uniform $U(a, b)$

$$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b, \text{ else } 0$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var} = \frac{(b-a)^2}{12}$$

7.2 Exponential $\text{Exp}(\lambda)$

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$F(x) = 1 - e^{-\lambda x}$$

$$P(X > t) = e^{-\lambda t}$$

$$E[X] = \frac{1}{\lambda}, \quad \text{Var} = \frac{1}{\lambda^2}$$

Memoryless:

$$P(X > s + t | X > s) = P(X > t)$$

Only continuous memoryless distribution.

MGF: $M(t) = \frac{\lambda}{\lambda - t}, t < \lambda$

7.3 Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

$$E[X] = \mu, \quad \text{Var} = \sigma^2$$

Standardize: $Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$

$$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

$$\Phi(-z) = 1 - \Phi(z)$$

$$P(|Z| \leq z) = 2\Phi(z) - 1$$

Linear transformation:

$$aX + b \sim N(a\mu + b, a^2\sigma^2)$$

Sum: $X \perp Y, X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2):$

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

MGF: $M(t) = \exp(\mu t + \sigma^2 t^2/2)$

8 Joint Distributions

8.1 Discrete

Joint PMF: $p_{X,Y}(x, y) = P(X = x, Y = y)$

Marginals:

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Conditional:

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

Independence:

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \text{ for all } x, y$$

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y)$$

8.2 Continuous

Joint PDF: $f_{X,Y}(x, y) \geq 0, \iint f dx dy = 1$

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

Marginals:

$$f_X(x) = \int f_{X,Y}(x, y) dy$$

Conditional:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Independence:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \text{ for all } x, y$$

If $f_{X,Y}(x, y)$ factorizes over rectangular domain, then $X \perp Y$.

$$E[g(X, Y)] = \iint g(x, y) f_{X,Y}(x, y) dx dy$$

8.3 Convolution

Discrete: $X \perp Y:$

$$P(X + Y = k) = \sum_j P(X = j)P(Y = k - j)$$

Continuous: $X \perp Y:$

$$f_{X+Y}(z) = \int f_X(x) f_Y(z - x) dx$$

8.4 Multinomial

n trials, k outcomes with probs $p_1, \dots, p_k$:

$$P(X_1 = n_1, \dots, X_k = n_k) = \frac{n!}{n_1! \dots n_k!} p_1^{n_1} \dots p_k^{n_k}$$

Marginals of multinomial are binomial.

9 Functions of RVs

9.1 Discrete

$$p_Y(y) = \sum_{x: g(x)=y} p_X(x)$$

$$\text{Indicator: } 1_A = \begin{cases} 1 & \omega \in A \\ 0 & \text{else} \end{cases}$$

$$E[1_A] = P(A)$$

9.2 Continuous

CDF method: $F_Y(y) = P(g(X) \leq y)$, then differentiate

Monotone transform: If $Y = g(X)$, g strictly monotone:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot |(g^{-1})'(y)|$$

10 Moment Generating Functions

$$M_X(t) = E[e^{tX}]$$

Properties:

- $E[X^k] = M_X^{(k)}(0)$ (k -th derivative at 0)
- MGF uniquely determines distribution
- $M_{aX+b}(t) = e^{bt} M_X(at)$
- $X \perp Y$: $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$

Common MGFs:

- Bern(p): $1 - p + pe^t$
- Bin(n, p): $(1 - p + pe^t)^n$
- Geom(p): $\frac{pe^t}{1 - (1-p)e^t}$
- Pois(λ): $e^{\lambda(e^t - 1)}$

11 Inequalities & Limit Theorems

11.1 Markov's Inequality

$X \geq 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

11.2 Chebyshev's Inequality

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

11.3 Weak Law of Large Numbers

i.i.d. X_i , $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

Proof:

$$P(|\bar{X} - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0$$

12 Useful Series

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \quad |r| < 1$$

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2}$$

$$\sum_{k=0}^{\infty} kr^k = \frac{r}{(1-r)^2}$$

$$\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^\lambda$$

13 Problem-Solving Techniques

1. **Indicator trick:** Write $X = \sum 1_{A_i}$, then $E[X] = \sum P(A_i)$. For Var, need independence.
2. **LOTP:** Partition sample space and sum: $P(A) = \sum P(A|B_i)P(B_i)$
3. **First-step analysis:** Condition on first step, write recursion for $P(A)$, solve.
4. **Tail sum:** $E[X] = \sum_{k \geq 1} P(X \geq k)$ for $X \in \{0, 1, \dots\}$
5. $E[X^2] = \text{Var} + \mu^2$: Always use this to recover second moments.
6. **Tower property:** $E[X] = E[E[X|Y]]$ to decompose by cases.
7. **MGF method:** Multiply MGFs of indep RVs, recognize distribution.
8. **Poisson approx:** n large, p small $\Rightarrow \text{Bin}(n, p) \approx \text{Pois}(np)$
9. **Sequential Bayes:** Use posterior as new prior.
10. **Complementary counting:** $P(A) = 1 - P(A^c)$

14 Common Exam Traps

- **Linearity of E needs NO independence; Var of sum DOES.**
- **Mutually exclusive $\neq$ independent.** If $A \cap B = \emptyset$ and $P(A), P(B) > 0$, then A, B are dependent.
- $P(A|B) + P(A|B^c) \neq 1$. Correct: $P(A|B) + P(A^c|B) = 1$
- $E[X^2] \neq (E[X])^2$ unless $\text{Var} = 0$
- **Geometric starts at $k = 1$; NegBin(r, p) at $k = r$**
- When probability depends on a random variable N : must condition on N using LOTP
- Always justify steps: name distributions, cite formulas, state independence
- Don't forget to divide by $P(\text{conditioning event})$
- $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$, but converse is FALSE
- Pairwise independence $\not\Rightarrow$ mutual independence
- In CDF method, be careful with inequalities: $P(g(X) \leq y)$ might need to flip depending on whether g is increasing or decreasing
- Memoryless property only holds for Geometric (discrete) and Exponential (continuous)
- Conditional probability: always check that conditioning event has positive probability
- MGF only works near $t = 0$; for some RVs it doesn't exist
- When using LOTUS, you don't need the PMF/PDF of $g(X)$ – use original distribution